

Dyadic-Sum Weighted Hankel Operators: Sharp Ideal Thresholds, 2-Adic Blocks, and Labeled Cycle Equations

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Abstract

We study the Dirichlet-weighted adjacency matrix supported on the dyadic anti-diagonals $m + n = 2^a$ of the positive-integer lattice. For $s \in \mathbb{C}$ and $\sigma = \Re s$, its matrix coefficients admit a bounded operator realization on $\ell^2(\mathbb{N})$ exactly when $\sigma > 0$; throughout that half-plane the operator is compact. We prove the sharp ideal walls $H_s \in \mathcal{S}_2$ exactly for $\sigma > 1/2$ and $H_s \in \mathcal{S}_1$ exactly for $\sigma > 1$. The arithmetic support also forces every edge to preserve the 2-adic valuation. Consequently, in the bounded region, H_s is the orthogonal direct sum of the scaled blocks $2^{-ks} A_s$, where A_s is the odd-vertex compression. This gives exact trace-power factors and a locally uniformly convergent block product for the Hilbert–Carleman determinant in the Hilbert–Schmidt half-plane. Separately, we solve every cyclic system $n_i + n_{i+1} = 2^{a_i}$ with fixed ordered labels: odd length has one candidate, whereas even length has an alternating compatibility condition followed by an explicit integer positivity interval. A canonical finite replay compared two independent implementations on four support cutoffs, 335,922 ordered label tuples, and 36 exact rational trace cases, with zero mismatches. These finite checks test implementation fidelity; every infinite statement and endpoint is established analytically.

1 Introduction

Arithmetic sparsity can make an infinite matrix simultaneously tractable and deceptive. In the matrix studied here, an entry is present only when its two indices sum to a power of two, but every present entry is weighted by both indices. The support is lacunary, the weights are not translation invariant, and the three basic operator-ideal questions have three different sharp answers.

For $s \in \mathbb{C}$, write $\sigma = \Re s$ and consider the coefficient array

$$h_s(m, n) = \mathbf{1}_{\{m+n=2^a \text{ for some } a \geq 1\}} (mn)^{-s/2}, \quad m, n \in \mathbb{N}, \quad (1.1)$$

where powers use the real logarithm and loops are retained. We prove that these coefficients define a bounded operator H_s on $\ell^2(\mathbb{N})$ exactly for $\sigma > 0$, and that the resulting operator is then compact. Its Hilbert–Schmidt and trace-class walls are, sharply,

$$H_s \in \mathcal{S}_2 \iff \sigma > \frac{1}{2}, \quad H_s \in \mathcal{S}_1 \iff \sigma > 1. \quad (1.2)$$

The endpoints are not consequences of a common interpolation argument. The wall at zero is detected by one infinite row, the Hilbert–Schmidt wall by central mass on each dyadic anti-diagonal, and the trace-class wall by pairwise disjoint trace-dual matchings.

The same support has an exact arithmetic symmetry. Every legal edge preserves the 2-adic valuation, so in the bounded half-plane the operator is an orthogonal direct sum of scaled copies of its odd-vertex block. This identity yields trace-power factors and, in the Hilbert–Schmidt

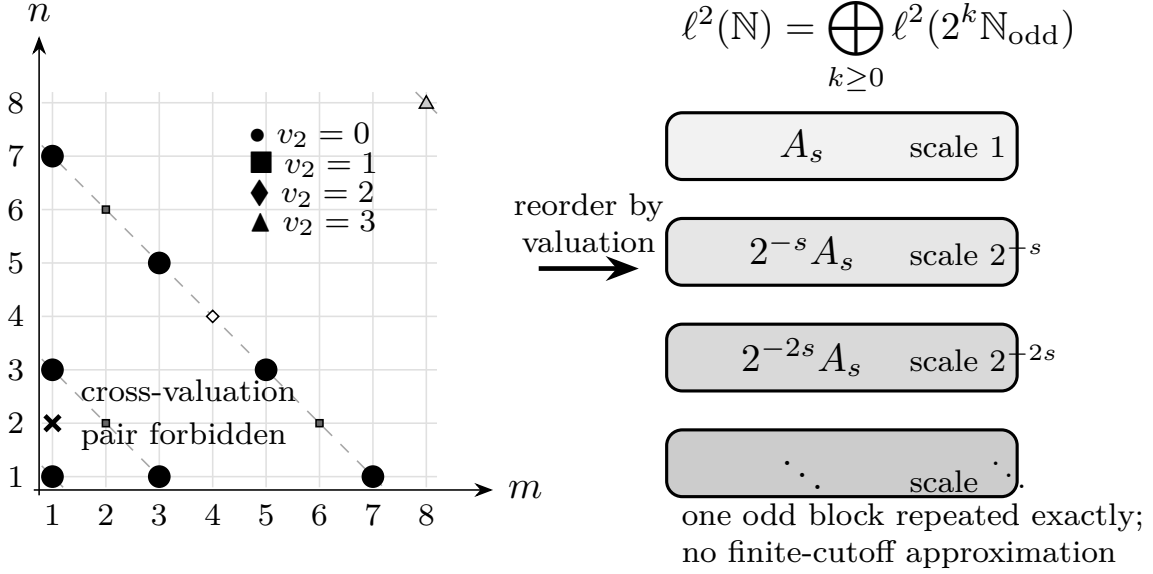


Figure 1: The support $m + n = 2^a$ lies on dyadic anti-diagonals, yet every legal edge preserves v_2 . Reordering the basis by exact valuation turns the bounded operator into scaled copies of its odd block. Marker shape and grayscale fill both identify the valuation, and the cross marks a candidate pair that is not an edge.

domain, a locally uniformly convergent block product for the Hilbert–Carleman determinant. It also explains why the marker for walk length and the 2-adic scale weight must remain different variables.

Our third component is algebraic rather than asymptotic. Given fixed ordered dyadic labels q_i , the equations $n_i + n_{i+1} = q_i$ around a cycle reduce to an alternating recurrence. Odd length fixes one candidate vertex word. Even length imposes one alternating compatibility equation and leaves an explicit finite interval for n_1 . This solver classifies based positive vertex solutions for the fixed ordered label tuple; it does not by itself quotient cyclic rotations or identify primitive orbits.

The contributions are therefore three precise statements.

1. We establish the sharp bounded/compact, Hilbert–Schmidt, and trace-class walls 0, $1/2$, and 1, including analytic endpoint obstructions.
2. We prove the exact bounded-operator decomposition $H_s \simeq \bigoplus_{k \geq 0} 2^{-ks} A_s$, derive its legal trace formulas, and give a self-contained proof of the regularized-determinant block product.
3. We give the complete odd/even solver for fixed ordered dyadic cycle labels and compare an algebraic implementation with a direct finite walk enumerator.

The computational comparison is deliberately narrower than the theorem. The exact finite replays agree without a mismatch on the declared grid; the complete counts are reported once in table 3. This agreement establishes implementation fidelity, not an infinite endpoint, external novelty, or a priority claim.

Section 2 positions the result. Section 3 fixes the source, types, and complex-parameter convention. Sections 4 and 5 prove the three walls. Section 6 develops valuation blocks, traces, and determinants, while section 7 solves the labeled cyclic systems. Section 8 reports the finite replay and closes with limitations.

2 Related work and ownership boundaries

Hankel operators and Schatten ideals. Classical Hankel operator theory relates ideal membership to analytic regularity and approximation properties; Peller’s characterization and monograph provide the standard setting and terminology [4, 5]. The matrix in equation (1.1) has Hankel *support*, but for $s \neq 0$ its values are not a function of $m + n$ alone: the left and right Dirichlet factors weight the two indices separately. Thus “weighted Hankel” in the title means a two-sided diagonal weighting of a Hankel-supported matrix, not a classical matrix with entries a_{m+n} . We use standard ideal facts, such as Hölder multiplication for Schatten classes, but prove the three thresholds directly from this particular coefficient array.

Lacunary support and Schur estimates. Fournier and Wagner develop Schur-test proofs in a lacunary Hankel setting and analyze reflection, folding, and alternating representations [2]. Those tools motivate the broader lacunary context. Our row estimate is reproduced in full because its two-sided Dirichlet weight is specific to equation (1.1); the generic Schur machinery remains attributed to that work.

Finite powers-of-two problems. Two nearby finite objects must be kept separate from the present infinite operator. Guo studies regularity of finite Hankel determinants formed from the characteristic sequence of the powers of two [3]. Alekseyev studies finite distinct-integer graph labelings whose endpoint sums are powers of two, together with associated restricted systems of linear equations and inequations [1]. Neither object is the Dirichlet-weighted operator on $\ell^2(\mathbb{N})$, and neither supplies its Schatten thresholds or determinant identities. Conversely, our cyclic solver fixes an ordered label tuple and allows repeated vertices and labels; it is not a solution of the finite distinct-label optimization problem.

Determinants. We use the Hilbert–Carleman determinant for Hilbert–Schmidt operators and the ordinary Fredholm determinant for trace-class operators; Simon gives the general trace-ideal framework [6]. The direct-sum identity needed below is proved inside the paper by combining the block eigenvalues and controlling the canonical factors. Thus the reference supplies definitions and background, not a page-level substitute for the specialized convergence argument.

A bounded search over these families did not locate the exact combined package of the three sharp walls, the valuation direct sum, and the ordered odd/even closure theorem. Together with the distinctions above, this records our claim and source boundary; it is not evidence that the package is first or unique.

3 The dyadic-sum source and its weighted operator

Let G_2 be the undirected looped graph with vertex set $\mathbb{N}$ and

$$m \sim n \iff m + n = 2^a \text{ for some integer } a \geq 1. \quad (3.1)$$

Its one-sided edge shift consists of sequences $(n_j)_{j \geq 0}$ satisfying $n_j \sim n_{j+1}$. One edge is one unit of time, and the formal marker z records that time. A primitive orbit, when needed, is a cyclic vertex word of least shift period, modulo rotation. Dyadic edge labels are constraints derived from a vertex word; they are not the primitive objects.

For $s \in \mathbb{C}$, define the coefficient array h_s by equation (1.1), with $n^{-s/2} = \exp[-(s/2) \log n]$ and the real logarithm. We say that h_s has a *bounded realization* if there is $H_s \in \mathcal{B}(\ell^2(\mathbb{N}))$ with matrix coefficients $\langle H_s e_n, e_m \rangle = h_s(m, n)$. This formulation matters when $\sigma \leq 0$: a column

Table 1: The typed objects used in the operator and walk ledgers.

Object	Mathematical type	Not identified with
n	positive-integer vertex	rational-prime primitive
$q_i = 2^{a_i}$	derived edge label	temporal orbit
$(n_1, \dots, n_r)$	based closed vertex walk	unordered labels
z	one-edge time marker	valuation weight
2^{-krs}	r -step block weight	time marker
H_s	bounded weighted adjacency, when legal	nonreal self-adjoint operator

then need not lie in ℓ^2 , so we do not silently introduce an unbounded operator on finitely supported vectors. On odd vertices, A_s denotes the corresponding compression whenever the bounded realization exists.

Lemma 3.1 (Left–right phase factorization). *Let $s = \sigma + it$, let $U_t e_n = n^{-it/2} e_n$, and write $H_s^{(N)} = (h_s(m, n))_{1 \leq m, n \leq N}$ and $U_t^{(N)} = \text{diag}(n^{-it/2})_{1 \leq n \leq N}$. Entrywise, and hence for every finite matrix,*

$$H_s^{(N)} = U_t^{(N)} H_\sigma^{(N)} U_t^{(N)}. \quad (3.2)$$

If the coefficient arrays have bounded realizations, then $H_s = U_t H_\sigma U_t$. Consequently boundedness, compactness, singular values, Schatten membership, and the corresponding norms depend only on σ .

Proof. The (m, n) coefficient on the right of equation (3.2) is $m^{-it/2} h_\sigma(m, n) n^{-it/2} = h_s(m, n)$. Each finite compression is an ordinary matrix identity. If H_σ is bounded, the same calculation gives the full bounded identity. Conversely, $U_t^* H_s U_t^*$ is a bounded operator with coefficients h_σ . Left and right multiplication by unitaries preserves singular values and all properties listed above. $\square$

The factorization is not a unitary conjugacy: the two copies of U_t point in the same direction. It therefore transfers neither spectra nor powers, traces, or determinants. All such identities below retain the complex parameter and come instead from an actual basis-reordering direct sum.

The loop locations and valuation constraint are elementary but structural. There is a loop at m precisely when $2m = 2^a$, hence precisely at $m = 2^k$, $k \geq 0$.

Lemma 3.2 (Valuation preservation). *If $m + n = 2^a$ with $m, n \in \mathbb{N}$, then $v_2(m) = v_2(n)$. If this common value is k and $m = 2^k u$, $n = 2^k v$, then u, v are odd and $u + v = 2^{a-k}$.*

Proof. If the two valuations differed, the valuation of their sum would equal the smaller one. Because $m, n < 2^a$, that value is strictly smaller than a , contradicting $v_2(m + n) = a$. Dividing the equality by 2^k gives the second statement. $\square$

4 The boundedness and compactness wall

The absolute row sums simultaneously give a Schur bound and a compactness criterion. For $\sigma > 0$, let

$$R_m = \sum_{n \geq 1} |h_s(m, n)|. \quad (4.1)$$

Let $A = A(m)$ be the least integer for which $2^A > m$. The neighbors of m are exactly $n_j = 2^{A+j} - m$, $j \geq 0$. The first neighbor is a positive integer, while for $j \geq 1$, $n_j \geq 2^{A+j-1}$.

Consequently the row sum satisfies

$$\begin{aligned} R_m &= m^{-\sigma/2} \sum_{j \geq 0} (2^{A+j} - m)^{-\sigma/2} \\ &\leq m^{-\sigma/2} + \frac{m^{-\sigma/2} 2^{-A\sigma/2}}{1 - 2^{-\sigma/2}} \leq m^{-\sigma/2} + C_\sigma m^{-\sigma}. \end{aligned} \quad (4.2)$$

In particular, $\sup_m R_m < \infty$ and $R_m \rightarrow 0$.

Theorem 4.1 (Sharp boundedness and compactness). *The coefficient array h_s admits a bounded operator realization on $\ell^2(\mathbb{N})$ if and only if $\sigma > 0$. Whenever it does, the realization H_s is compact.*

Proof. For $\sigma > 0$, the modulus array is symmetric, so the Schur test and equation (4.2) give a bounded realization. To prove compactness, fix $\varepsilon > 0$. Choose M so that $R_m < \varepsilon$ for $m > M$. Each of the finitely many rows $m \leq M$ is absolutely summable, so some $N \geq M$ satisfies

$$\sum_{n > N} |h_s(m, n)| < \varepsilon, \quad 1 \leq m \leq M.$$

Every absolute row sum of $H_s - P_N H_s P_N$ is then below ε : for $m \leq M$ it is the selected row tail, and for $m > M$ it is at most R_m . The same holds for columns by symmetry of the modulus array. A second Schur estimate yields $\|H_s - P_N H_s P_N\| \leq \varepsilon$. Hence finite-rank compressions converge in norm.

Suppose now that $\sigma \leq 0$ and that a bounded operator had the declared matrix coefficients. In its row indexed by 1, the entries at $n = 2^a - 1$, $a \geq 1$, have squared moduli $(2^a - 1)^{-\sigma}$. Therefore

$$\sum_{a \geq 1} |h_s(1, 2^a - 1)|^2 = \sum_{a \geq 1} (2^a - 1)^{-\sigma} = \infty. \quad (4.3)$$

But a row of a bounded operator is the coordinate sequence of its adjoint applied to a basis vector and must lie in ℓ^2 . This contradiction excludes every bounded realization on $\sigma \leq 0$. $\square$

The proof is genuinely infinite at the boundary: no finite singular-value calculation can witness the divergent row in equation (4.3).

5 Sharp Hilbert–Schmidt and trace-class thresholds

Distinct dyadic anti-diagonals have disjoint support. Hence, whenever the sum is finite,

$$\|H_s\|_2^2 = \sum_{a \geq 1} \sum_{m=1}^{2^a-1} [m(2^a - m)]^{-\sigma}. \quad (5.1)$$

The following elementary level estimates contain the entire Hilbert–Schmidt threshold. Their detailed constants are recorded in appendix A.

Lemma 5.1 (Dyadic anti-diagonal estimates). *For $N = 2^a$:*

1. *if $0 < \alpha < 1$, then*

$$\sum_{m=1}^{N-1} [m(N - m)]^{-\alpha} \asymp_\alpha N^{1-2\alpha};$$

2. *if $\alpha = 1$, then the sum is $2H_{N-1}/N$;*

3. *if $\alpha > 1$, then the sum is $O_\alpha(N^{-\alpha})$.*

Proof. By symmetry it suffices to sum over $m \leq N/2$. There, $N/2 \leq N - m < N$, so the level is comparable to $N^{-\alpha} \sum_{m \leq N/2} m^{-\alpha}$. This is $N^{1-2\alpha}$ for $\alpha < 1$ and $O(N^{-\alpha})$ for $\alpha > 1$. At $\alpha = 1$, the partial-fraction identity

$$\frac{1}{m(N-m)} = \frac{1}{N} \left(\frac{1}{m} + \frac{1}{N-m} \right)$$

gives the exact formula. $\square$

Theorem 5.2 (Sharp Hilbert–Schmidt wall). *The bounded realization H_s belongs to $\mathcal{S}_2$ if and only if $\sigma > 1/2$.*

Proof. For $0 < \sigma < 1$, applying the first part of lemma 5.1 to equation (5.1) gives a dyadic outer series comparable to $\sum_a 2^{a(1-2\sigma)}$, which converges exactly when $\sigma > 1/2$. In particular, the level mass stays bounded below at $\sigma = 1/2$. At $\sigma = 1$, the exact harmonic formula is $2H_{2^a-1}/2^a$, which is summable in a . For $\sigma > 1$, the third part gives an $O(2^{-a\sigma})$ level bound. Finally, theorem 4.1 already excludes $\sigma \leq 0$. $\square$

Trace class needs a different endpoint witness. Sufficiency follows from entrywise summability:

$$\sum_{m,n \geq 1} |h_s(m,n)| = \sum_{a \geq 1} \sum_{m=1}^{2^a-1} [m(2^a - m)]^{-\sigma/2}. \quad (5.2)$$

For $1 < \sigma < 2$, the a th level is $O(2^{a(1-\sigma)})$; for $\sigma = 2$ it is $2H_{2^a-1}/2^a = O(a2^{-a})$; and for $\sigma > 2$ it is $O(2^{-a\sigma/2})$. These level bounds make equation (5.2) finite whenever $\sigma > 1$. Expanding the matrix into rank-one coordinate operators then converges absolutely in trace norm.

For necessity, set $Q_j = 4^j$ and

$$I_j = \mathbb{Z} \cap [Q_j/4, Q_j/3], \quad J_j = \{Q_j - m : m \in I_j\}. \quad (5.3)$$

All sets I_j and J_j are pairwise disjoint. For a finite upper index J , define a partial isometry V_J by

$$V_J e_{Q_j-m} = \frac{h_s(m, Q_j-m)}{|h_s(m, Q_j-m)|} e_m, \quad m \in I_j, \quad j \leq J,$$

and let it vanish on the orthogonal complement. Its selected domain and range vectors are orthonormal, so $\|V_J\| = 1$. Direct matrix multiplication gives

$$\mathrm{Tr}(V_J^* H_s) = \sum_{j \leq J} \sum_{m \in I_j} [m(Q_j - m)]^{-\sigma/2}. \quad (5.4)$$

Each inner sum is comparable to $Q_j^{1-\sigma}$: it contains a fixed positive proportion of Q_j indices, and both factors are comparable to Q_j . If H_s were trace class for $\sigma \leq 1$, standard trace duality [6] would bound equation (5.4) by $\|H_s\|_1$ uniformly in J , whereas its right-hand side diverges.

Theorem 5.3 (Sharp trace-class wall). *The bounded realization H_s belongs to $\mathcal{S}_1$ if and only if $\sigma > 1$.*

Proof. Entrywise summability proves sufficiency. The disjoint matching argument proves necessity for $0 < \sigma \leq 1$, including the endpoint; for $\sigma \leq 0$ there is no bounded realization by theorem 4.1. $\square$

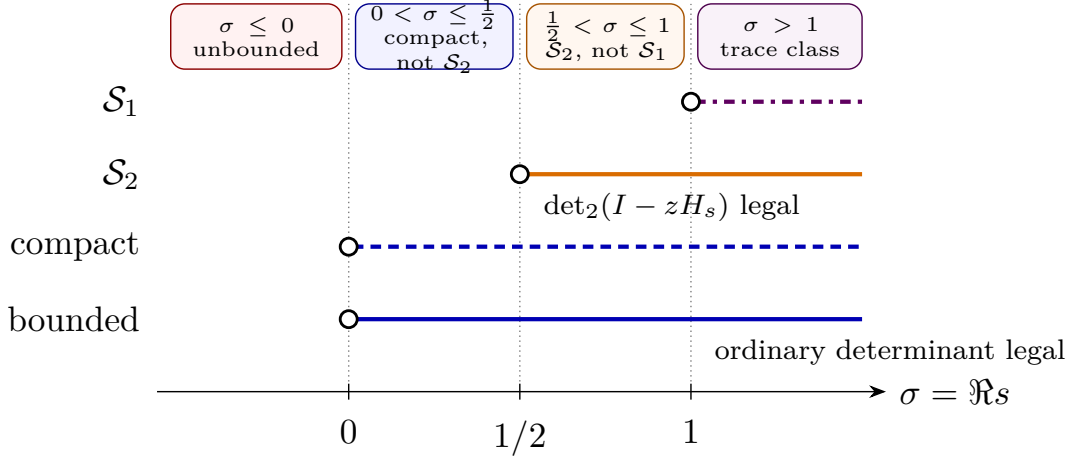


Figure 2: The three strict walls require different analytic witnesses. Open circles exclude the endpoints: boundedness and compactness begin at $\sigma > 0$, the Hilbert–Schmidt property at $\sigma > 1/2$, and trace class at $\sigma > 1$. Line style duplicates the color encoding.

Table 2: Exact operator-ideal and determinant domains. Every endpoint is rejected analytically rather than by finite-cutoff extrapolation.

Property	Exact domain	Boundary obstruction
Bounded and compact	$\sigma > 0$	row $m = 1$ is not in ℓ^2
Hilbert–Schmidt	$\sigma > \frac{1}{2}$	central anti-diagonal mass
Trace class	$\sigma > 1$	disjoint trace-dual matchings
$\det_2(I - zH_s)$	$\sigma > \frac{1}{2}$	requires $H_s \in \mathcal{S}_2$
$\det(I - zH_s)$	$\sigma > 1$	requires $H_s \in \mathcal{S}_1$

6 Valuation blocks, trace powers, and legal determinants

For $k \geq 0$, let

$$\mathcal{H}_k = \overline{\text{span}}\{e_{2^k u} : u \in \mathbb{N}_{\text{odd}}\}.$$

These spaces are mutually orthogonal and exhaust $\ell^2(\mathbb{N})$. Let $W_k : \ell^2(\mathbb{N}_{\text{odd}}) \rightarrow \mathcal{H}_k$ send e_u to $e_{2^k u}$. Define

$$\mathcal{K} := \bigoplus_{k \geq 0} \ell^2(\mathbb{N}_{\text{odd}}), \quad W : \mathcal{K} \longrightarrow \ell^2(\mathbb{N}), \quad W((x_k)_{k \geq 0}) = \sum_{k \geq 0} W_k x_k.$$

Orthogonality of the $\mathcal{H}_k$ makes W a unitary. Every block direct sum below is an operator on the explicitly named source space $\mathcal{K}$.

Theorem 6.1 (Exact 2-adic direct sum). *If $\sigma > 0$, then, as bounded operators,*

$$W^* H_s W = \bigoplus_{k \geq 0} 2^{-ks} A_s. \tag{6.1}$$

No unbounded-operator equivalence is asserted for $\sigma \leq 0$.

Proof. Lemma 3.2 makes all off-diagonal blocks between distinct $\mathcal{H}_k$ vanish. If $m = 2^k u$ and $n = 2^k v$ with u, v odd, then

$$h_s(m, n) = 2^{-ks} \mathbf{1}_{\{u+v=2^b \text{ for some } b \geq 1\}} (uv)^{-s/2}.$$

Thus the k th diagonal block is exactly $2^{-ks} A_s$. Boundedness on $\sigma > 0$ is supplied by theorem 4.1, so the block identity is an identity of bounded operators. $\square$

The direct sum now carries operator algebra, unlike the left–right phase factorization in lemma 3.1. In the Hilbert–Schmidt half-plane, H_s^2 and A_s^2 are trace class by Schatten Hölder, and multiplication by bounded powers handles every $r \geq 2$.

Proposition 6.2 (Trace powers and the ordinary trace). *If $\sigma > 1/2$ and $r \geq 2$, then*

$$\mathrm{Tr}(H_s^r) = \frac{\mathrm{Tr}(A_s^r)}{1 - 2^{-rs}}. \quad (6.2)$$

If $\sigma > 1$, then

$$\mathrm{Tr}(H_s) = \sum_{k \geq 0} 2^{-ks} = \frac{1}{1 - 2^{-s}}. \quad (6.3)$$

Proof. Taking the r th power of equation (6.1) gives blocks $2^{-krs}A_s^r$. Because the full power is trace class, its trace is the absolutely convergent sum of the block traces. The scalar geometric series converges since $r\sigma > 1$, proving equation (6.2). In the trace-class domain, the trace is the absolutely convergent diagonal sum. Loops occur exactly at 2^k , and the diagonal entry there is 2^{-ks} , which gives equation (6.3). $\square$

We next separate an entire determinant identity from its local logarithm. For a Hilbert–Schmidt operator T , write

$$E_2(w) = (1 - w)e^w, \quad \det_2(I - zT) = \prod_j E_2(z\lambda_j(T)),$$

where eigenvalues are repeated with algebraic multiplicity. This is the standard Hilbert–Carleman determinant convention [6].

Proposition 6.3 (Regularized block product). *If $\sigma > 1/2$, then*

$$\det_2(I - zH_s) = \prod_{k \geq 0} \det_2(I - z2^{-ks}A_s), \quad (6.4)$$

and the product converges uniformly on every compact subset of $\mathbb{C}$.

Proof. Set $T_k = 2^{-ks}A_s$ on the k th summand of $\mathcal{K}$. Each T_k is Hilbert–Schmidt, and

$$\sum_{k \geq 0} \|T_k\|_2^2 = \|A_s\|_2^2 \sum_{k \geq 0} 2^{-2k\sigma} < \infty.$$

By theorem 6.1, $\bigoplus_k T_k = W^*H_sW$. Applying the square-summable direct-sum theorem lemma B.1 therefore gives equation (6.4) and compact-uniform convergence. That lemma also supplies the complete nonnormal combined-eigenvalue argument, including the finite exceptional factors and canonical-product tail control. $\square$

Both sides of equation (6.4) are entire in z . On the smaller zero-free disk $|z|\|H_s\| < 1$, choose the logarithm branch normalized by $\log \det_2(I) = 0$. Then

$$\log \det_2(I - zH_s) = - \sum_{r \geq 2} \frac{z^r}{r} \frac{\mathrm{Tr}(A_s^r)}{1 - 2^{-rs}}. \quad (6.5)$$

This logarithmic series is local; it is not a global logarithm of the entire function in equation (6.4).

In the overlap $\sigma > 1$, H_s and A_s are trace class. Let $D_K = \bigoplus_{k=0}^K 2^{-ks}A_s$. Then

$$\|W^*H_sW - D_K\|_1 = \|A_s\|_1 \sum_{k > K} 2^{-k\sigma} \rightarrow 0.$$

For finite K , the Fredholm determinant of $I - zD_K$ is the product of the block determinants. Continuity of the determinant in trace norm, uniformly for z in a compact set, permits $K \rightarrow \infty$ and gives

$$\det(I - zH_s) = \prod_{k \geq 0} \det(I - z2^{-ks}A_s) \quad (6.6)$$

locally uniformly in z . The two determinant conventions agree through

$$\det_2(I - zH_s) = \det(I - zH_s) \exp\left(\frac{z}{1 - 2^{-s}}\right). \quad (6.7)$$

No ordinary determinant is used on $1/2 < \sigma \leq 1$, and no infinite determinant is defined outside its corresponding ideal domain.

7 Complete labeled cyclic closure

Fix a length $r \geq 1$ and an ordered tuple of dyadic labels $q_i = 2^{a_i} \geq 2$. We seek positive integers $n_1, \dots, n_r$, with $n_{r+1} = n_1$, satisfying

$$n_i + n_{i+1} = q_i, \quad 1 \leq i \leq r. \quad (7.1)$$

The length range here includes $r = 1$ because the graph retains loops. By contrast, traces in the Hilbert–Schmidt-only domain use $r \geq 2$.

Iterating $n_{i+1} = q_i - n_i$ gives, for $2 \leq i \leq r + 1$,

$$n_i = (-1)^{i-1}n_1 + \sum_{j=1}^{i-1} (-1)^{i-1-j}q_j. \quad (7.2)$$

Closing at $i = r + 1$ yields the single equation

$$(1 - (-1)^r)n_1 = \sum_{j=1}^r (-1)^{r-j}q_j. \quad (7.3)$$

Theorem 7.1 (Odd/even labeled cycle solver). *For the ordered system equation (7.1):*

1. *If r is odd, there is one algebraic candidate,*

$$n_1 = \frac{1}{2}(q_1 - q_2 + q_3 - \dots + q_r). \quad (7.4)$$

It is a positive solution exactly when every vertex obtained from equation (7.2) is positive.

2. *If r is even, a solution exists only if*

$$q_1 - q_2 + q_3 - \dots - q_r = 0. \quad (7.5)$$

Under this condition, set

$$b_1 = 0, \quad b_i = \sum_{j=1}^{i-1} (-1)^{i-1-j}q_j \quad (2 \leq i \leq r), \quad (7.6)$$

and

$$L(q) = \max_{\substack{1 \leq i \leq r \\ i \text{ odd}}} (-b_i), \quad U(q) = \min_{\substack{1 \leq i \leq r \\ i \text{ even}}} b_i. \quad (7.7)$$

The positive solutions are exactly

$$n_1 = x \in \mathbb{Z} \cap (L(q), U(q)), \quad n_i = (-1)^{i-1}x + b_i. \quad (7.8)$$

Thus their number is $\max\{0, U(q) - L(q) - 1\}$.

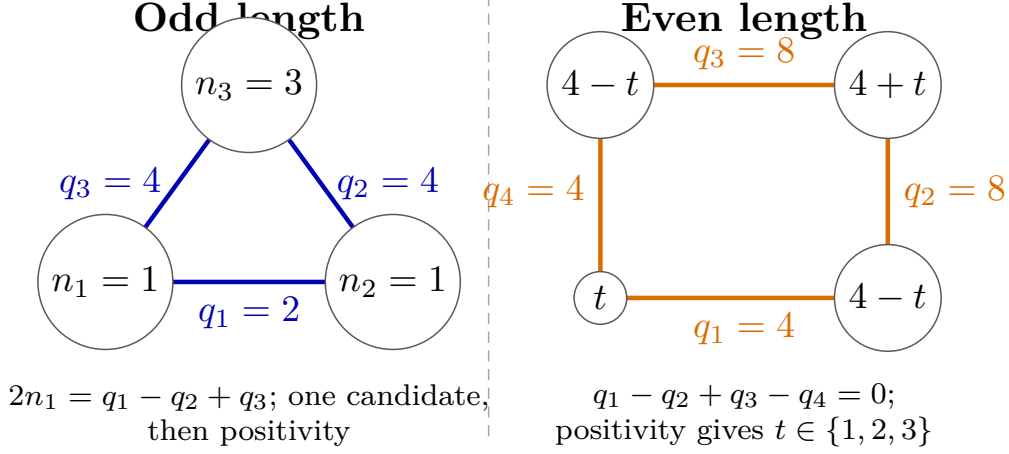


Figure 3: For fixed ordered dyadic labels, odd length fixes one candidate vertex word, whereas even length first imposes the alternating compatibility condition and then leaves an integer positivity interval. The displayed examples classify based vertex solutions, not primitive orbits modulo rotation.

In the odd block A_s , retain precisely those solutions for which n_1 is odd.

Proof. Equations (7.2) and (7.3) follow by induction. If r is odd, the coefficient on the left of equation (7.3) is 2, which gives equation (7.4); it is integral because every q_i is even. The recurrence fixes every remaining vertex, so positivity is the only condition left.

If r is even, the left side of equation (7.3) vanishes, and its right side vanishes exactly under equation (7.5). Write $x = n_1$. Formula equation (7.2) becomes $n_i = (-1)^{i-1}x + b_i$. For odd i , positivity says $x > -b_i$; for even i , it says $x < b_i$. Their intersection is exactly $\mathbb{Z} \cap (L(q), U(q))$. The endpoints are integers, giving the stated count. Finally, subtracting a vertex from the even integer q_i preserves parity along every edge. Hence all vertices are odd exactly when n_1 is odd. $\square$

The frozen examples illustrate all branches. Labels $(2, 4, 4)$ give the unique odd-length solution $(1, 1, 3)$. Labels $(4, 8, 8, 4)$ satisfy equation (7.5); here

$$(n_1, n_2, n_3, n_4) = (x, 4 - x, 4 + x, 4 - x),$$

so positivity gives $x \in \{1, 2, 3\}$, with $x = 1, 3$ in the odd block. Labels $(4, 4, 8, 4)$ have nonzero alternating sum and no solution.

The passage from matrices to an absolutely convergent infinite walk sum uses the following ideal estimate.

Lemma 7.2 (Powers of finite compressions). *Let $T \in \mathcal{S}_2$, let P_N be the coordinate projection onto the first N basis vectors, and set $T_N = P_N T P_N$. Then $\|T_N - T\|_2 \rightarrow 0$ and, for every integer $r \geq 2$,*

$$\|T_N^r - T^r\|_1 \rightarrow 0. \quad (7.9)$$

Proof. The Hilbert–Schmidt convergence follows first for finite-rank operators and then for T by finite-rank density, because left and right multiplication by an orthogonal projection are contractions in $\mathcal{S}_2$. Put $\delta_N = \|T_N - T\|_2$. Schatten Hölder gives the explicit square estimate

$$\|T_N^2 - T^2\|_1 \leq \delta_N (\|T_N\|_2 + \|T\|_2) \rightarrow 0. \quad (7.10)$$

For $r > 2$, the exact identity

$$T_N^r - T^r = (T_N^2 - T^2)T_N^{r-2} + T^2(T_N^{r-2} - T^{r-2})$$

and the ideal inequality yield

$$\|T_N^r - T^r\|_1 \leq \|T_N^2 - T^2\|_1 \|T_N\|^{r-2} + \|T^2\|_1 \|T_N^{r-2} - T^{r-2}\|.$$

Here $\|T_N\| \leq \|T\|$ and $\|T_N - T\| \leq \delta_N$; the usual telescoping formula for bounded powers therefore makes the last factor tend to zero. This proves equation (7.9). $\square$

When $\sigma > 1/2$ and $r \geq 2$, the trace admits a closed-walk interpretation. The product of edge weights around a based closed walk is

$$\prod_{i=1}^r h_s(n_i, n_{i+1}) = \prod_{i=1}^r n_i^{-s}. \quad (7.11)$$

Indeed, apply lemma 7.2 first to H_σ . For every N , expansion of the finite matrix trace gives

$$\text{Tr}((P_N H_\sigma P_N)^r) = \sum_{1 \leq n_1, \dots, n_r \leq N} \prod_{i=1}^r h_\sigma(n_i, n_{i+1}), \quad n_{r+1} = n_1.$$

Every summand is nonnegative. These finite sums increase with N and, by equation (7.9), converge to the finite value $\text{Tr}(H_\sigma^r)$. Moreover,

$$\left| \prod_{i=1}^r h_s(n_i, n_{i+1}) \right| = \prod_{i=1}^r h_\sigma(n_i, n_{i+1}),$$

so the complex closed-walk series is absolutely convergent by comparison. A second application of the lemma to H_s identifies its sum with $\text{Tr}(H_s^r)$. Consequently the solver is a matrix-independent trace evaluator: enumerate an ordered label tuple, solve it using theorem 7.1, and add equation (7.11) for the accepted based vertex words. Primitive-period reduction and quotienting by cyclic base point are separate operations, as detailed in appendix C.

8 Independent replay, limitations, and conclusion

8.1 Canonical finite replay

The replay compares two structurally independent finite representations. Evaluator M constructs cutoff matrices from the bit predicate and uses matrix operations; Evaluator C constructs dyadic anti-diagonals, partitions them by valuation, and applies theorem 7.1 without constructing Evaluator M's matrix.

Table 3: Canonical State-A implementation replay. These finite exact comparisons test two independent implementations; they do not prove an infinite theorem or a priority claim.

Replay surface	Cases	Mismatches
Complete support cutoffs	4	0
Ordered dyadic label tuples	335,922	0
Exact rational finite traces	36	0

The table covers complete cutoffs $N = 8, 16, 32, 64$, every ordered label tuple in the declared grid, and 36 exact rational traces. All comparisons use strict recursive type-and-value equality. Finite traces retain their scale-dependent odd-block cutoffs and endpoint partial sums remain finite diagnostics. Thus the replay tests implementation fidelity only; the infinite statements are proved analytically. Exact truncation, hash, integrity, and adversarial metadata are recorded in appendix D.

8.2 Limitations and conclusion

The results concern one frozen looped graph, one real-logarithm branch, and one Dirichlet weighting. We prove neither an all- $\mathcal{S}_p$ theorem nor any connection between this powers-of-two support and rational primes; no Hilbert–Polya operator, completed divisor, or functional equation is claimed. For nonreal s , H_s is complex symmetric but is not asserted to be Hermitian. The cycle theorem classifies a fixed ordered label system; it does not by itself enumerate primitive orbits modulo rotation.

Within those boundaries, one dyadic constraint controls both halves of the paper. Its anti-diagonal geometry creates the strict ideal walls, while its valuation invariance turns the bounded operator into repeated odd blocks. The alternating recurrence is the finite-dimensional counterpart of that same support and closes every labeled cyclic system. A concrete next question is whether the intermediate Schatten classes of this same frozen operator admit comparably sharp thresholds; the cases $p = 1, 2$ proved here do not justify extrapolating an answer.

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A Endpoint estimates and the trace-dual matching

This appendix records the quantitative bounds used in sections 4 and 5. They are included to make the endpoint strictness independent of any finite-cutoff trend.

A.1 Anti-diagonal levels

For $N \geq 2$ and $\alpha > 0$, put

$$S_\alpha(N) = \sum_{m=1}^{N-1} [m(N-m)]^{-\alpha}.$$

By symmetry,

$$S_\alpha(N) = 2 \sum_{1 \leq m < N/2} [m(N-m)]^{-\alpha} + \mathbf{1}_{\{N \text{ even}\}} (N^2/4)^{-\alpha}.$$

On $m \leq N/2$,

$$N^{-\alpha} m^{-\alpha} \leq [m(N-m)]^{-\alpha} \leq 2^\alpha N^{-\alpha} m^{-\alpha}.$$

For $0 < \alpha < 1$, the integral comparison

$$c_\alpha M^{1-\alpha} \leq \sum_{m \leq M} m^{-\alpha} \leq C_\alpha M^{1-\alpha}$$

therefore gives

$$c_\alpha N^{1-2\alpha} \leq S_\alpha(N) \leq C_\alpha N^{1-2\alpha}.$$

At $\alpha = 1$, summing the partial-fraction identity in lemma 5.1 gives $S_1(N) = 2H_{N-1}/N$. For $\alpha > 1$, the same half-interval estimate and $\sum_{m \geq 1} m^{-\alpha} < \infty$ give $S_\alpha(N) \leq C_\alpha N^{-\alpha}$.

At the Hilbert–Schmidt boundary $\sigma = 1/2$, one can see divergence without invoking the full comparison. For $N/4 \leq m \leq 3N/4$, both factors are at most $3N/4$, so each of the $\gg N$ central summands in equation (5.1) is $\gg N^{-1}$. Every dyadic level therefore has mass bounded below by a positive constant.

A.2 Compactness from row decay

The row-decay estimate in equation (4.2) gives both $\sup_m R_m < \infty$ and $R_m \rightarrow 0$. The second assertion alone does not immediately make an arbitrary bounded matrix compact, so we record the finite-row tail step. For fixed M , absolute summability of each of the rows $1, \dots, M$ permits one common N for their tails. Rows above M are controlled in their entirety by R_m . Thus the modulus of $H_s - P_N H_s P_N$ has uniformly small row and column sums. This is the precise norm-approximation argument used in theorem 4.1.

A.3 Disjoint matching details

For $Q_j = 4^j$, the sets in equation (5.3) satisfy

$$I_j \subset [Q_j/4, Q_j/3], \quad J_j \subset [2Q_j/3, 3Q_j/4].$$

They are disjoint within a scale. Their union lies below $3Q_j/4$, whereas the next interval I_{j+1} begins at $Q_{j+1}/4 = Q_j$; consequently all domains and ranges used by the matchings are pairwise disjoint across scales.

If $m \in I_j$ and $n \in J_j$, then $11Q_j/12 \leq m + n \leq 13Q_j/12$. The only power of two in that interval is Q_j , so the selected support entry is present exactly when $n = Q_j - m$. Define the matrix of V_J on these entries by

$$(V_J)_{m, Q_j - m} = \frac{h_s(m, Q_j - m)}{|h_s(m, Q_j - m)|}.$$

Disjointness makes V_J a partial isometry of norm one. With the matrix convention $T_{m,n} = \langle T e_n, e_m \rangle$,

$$\text{Tr}(V_J^* H_s) = \sum_{j \leq J} \sum_{m \in I_j} \overline{(V_J)_{m, Q_j - m}} h_s(m, Q_j - m),$$

which is the positive sum in equation (5.4). Moreover, $|I_j| \asymp Q_j$ and $m, Q_j - m \asymp Q_j$ uniformly on I_j , so

$$\sum_{m \in I_j} [m(Q_j - m)]^{-\sigma/2} \asymp_\sigma Q_j^{1-\sigma}.$$

For $\sigma = 1$ each scale contributes a positive constant; for $\sigma < 1$ its contribution grows. Trace duality would require all finite partial sums to be at most $\|H_s\|_1$, completing the endpoint contradiction without estimating finite-cutoff trace norms.

B Direct sums of Hilbert–Carleman determinants

We prove the specialized direct-sum statement used in proposition 6.3. The proof is included so that the block product does not depend on a brittle page citation.

Lemma B.1 (Square-summable direct-sum product). *Let T_k be compact Hilbert–Schmidt operators, and suppose*

$$\sum_{k \geq 0} \|T_k\|_2^2 < \infty.$$

Then $T = \bigoplus_{k \geq 0} T_k$ is Hilbert–Schmidt and

$$\det_2(I - zT) = \prod_{k \geq 0} \det_2(I - zT_k), \quad (\text{B.1})$$

with uniform convergence on compact subsets of $\mathbb{C}$.

Proof. List the nonzero eigenvalues of T_k , with algebraic multiplicity, as $\lambda_{k,j}$. Weyl’s eigenvalue inequality gives

$$\sum_{k,j} |\lambda_{k,j}|^2 \leq \sum_k \|T_k\|_2^2 < \infty. \quad (\text{B.2})$$

Also $\|T_k\| \leq \|T_k\|_2 \rightarrow 0$. Hence the nonzero eigenvalues of the compact direct sum T are exactly the combined multiset $\{\lambda_{k,j}\}_{k,j}$.

Fix a compact set $K \subset \mathbb{C}$ and put $R = \max_{z \in K} |z|$. By equation (B.2), only finitely many combined eigenvalues satisfy $R|\lambda_{k,j}| > 1/2$. For every remaining factor, use the analytic branch at zero:

$$\log E_2(w) = \log(1 - w) + w = - \sum_{r \geq 2} \frac{w^r}{r}, \quad |w| \leq \frac{1}{2}. \quad (\text{B.3})$$

The series gives $|\log E_2(w)| \leq |w|^2$ on this disk. Therefore

$$\sum_{k,j} \sup_{z \in K} |\log E_2(z\lambda_{k,j})| \leq R^2 \sum_{k,j} |\lambda_{k,j}|^2 < \infty$$

after removing the finite exceptional set. The combined canonical product thus converges normally on K . Normal convergence makes its value independent of the ordering and permits grouping first over j and then over k . The ungrouped product is the eigenvalue definition of $\det_2(I - zT)$; each grouped product is $\det_2(I - zT_k)$. This proves equation (B.1), including compact-uniform convergence. Zeros among the finitely many exceptional factors cause no difficulty, because the identity follows between the resulting entire products. $\square$

Apply the lemma with $T_k = 2^{-ks} A_s$. In the domain $\sigma > 1/2$, $A_s \in \mathcal{S}_2$ and

$$\sum_{k \geq 0} \|T_k\|_2^2 = \|A_s\|_2^2 \sum_{k \geq 0} 2^{-2k\sigma} < \infty.$$

The unitary equivalence in theorem 6.1 identifies $T = \bigoplus_k T_k$ with H_s , proving equation (6.4).

For completeness, if $|z| \|H_s\| < 1$, then $I - zH_s$ is invertible by the Neumann series, so the determinant has no zero on that disk. The branch of its logarithm fixed at zero satisfies

$$\log \det_2(I - zH_s) = - \sum_{r \geq 2} \frac{z^r}{r} \operatorname{Tr}(H_s^r).$$

Indeed,

$$|\operatorname{Tr}(H_s^r)| \leq \|H_s^r\|_1 \leq \|H_s\|_2^2 \|H_s\|^{r-2},$$

so the series converges absolutely under the displayed condition. Inserting equation (6.2) gives equation (6.5). This derivation also shows why the local logarithm cannot be promoted to a single global branch when the entire determinant has zeros.

If $\sigma > 1$, then $A_s \in \mathcal{S}_1$ and

$$\sum_{k \geq 0} \|2^{-ks} A_s\|_1 = \|A_s\|_1 \sum_{k \geq 0} 2^{-k\sigma} < \infty.$$

The genus-zero eigenvalue products consequently converge normally on compact z -sets and give equation (6.6). Finally, the defining relation $\det_2(I + T) = \det(I + T)e^{-\operatorname{Tr} T}$ for trace-class T , applied to $T = -zH_s$ and combined with equation (6.3), gives equation (6.7).

C Cycle intervals, parity, and orbit bookkeeping

The affine coordinates in theorem 7.1 make positivity transparent. Under even compatibility,

$$n_i = (-1)^{i-1} x + b_i.$$

Every odd index contributes a strict lower bound $x > -b_i$, and every even index contributes a strict upper bound $x < b_i$. Both index sets are nonempty for even $r \geq 2$, so the interval $(L(q), U(q))$ is finite. It may be empty. Since all q_i and hence all b_i are integers, its positive solution count is exactly

$$\#(\mathbb{Z} \cap (L(q), U(q))) = \max\{0, U(q) - L(q) - 1\}.$$

For the odd block, the recurrence modulo two reads $n_{i+1} \equiv n_i \pmod{2}$, because q_i is even. Thus filtering $x = n_1$ to odd values filters every vertex simultaneously.

At $r = 1$, equation (7.4) gives $n_1 = q_1/2$. This is exactly the loop at a power of two. In the odd block only $q_1 = 2$ remains, corresponding to the loop at vertex 1. This length-one convention is independent of the restriction $r \geq 2$ used for Hilbert–Schmidt trace powers.

Based words, rotations, and primitive period. A solution of equation (7.1) is a based vertex word $(n_1, \dots, n_r)$. Rotating the word changes its base point but represents the same cyclic orbit. The edge labels rotate with it. A word is primitive only when its least cyclic shift period is r ; a repeated traversal is not a new primitive orbit. The label tuple is derived from the vertex word and does not replace it: the same compatible even label tuple can have several vertex solutions, as $(4, 8, 8, 4)$ demonstrates.

The coefficient $1/r$ in the local determinant logarithm removes cyclic base points at the level of based trace terms. It does not declare every label tuple primitive, nor does it turn a repeated word into a new primitive. Any Euler-product reorganization would require a separate least-period reduction. No such primitive-orbit product is claimed in this paper.

Finite-cutoff evaluation. For a cutoff N , a matrix-independent evaluator lists the allowed ordered labels, applies theorem 7.1, and then retains precisely the solutions with $1 \leq n_i \leq N$ for every i . Summing $\prod_i n_i^{-s}$ over these based solutions agrees with the trace of the cutoff matrix power. The procedure is exact because the theorem classifies all positive solutions for each fixed ordered tuple; it uses neither a spectral approximation nor a primitive-orbit heuristic.

D Canonical State-A evidence binding

The manuscript table was generated from a protected State-A snapshot by a writer-side extractor. The extractor rejects duplicate JSON keys, noncanonical serialization, unexpected paths, failed statuses, and any ledger/hash mismatch before emitting the local summary. The relevant bindings are listed below.

State-A result ledger. `fa22dde6ec3a9cbd473528ebb619863ac7beb0d1c9cc807394541501153add37.`

Science projection. `8f03cf68855e1614b368d8f464349b4f96cc6ab1ac0faaddaa35c748e63080de.`

Writer canonical summary. `c86887d3e7e9602cfebaec3e0b03e534d243af576166115fd7825f130a8ec774.`

Protected authority snapshot. `ca9f33594949f405e3df72404d35ebd4d26d8fd1f314d7be940367709ed905e8.`

The exact finite block identity retains a different odd cutoff at every valuation scale. If $A_s^{(M)}$ denotes the odd block restricted to odd $u \leq M$, then the evaluated identity is

$$\mathrm{Tr}((P_N H_s P_N)^r) = \sum_{0 \leq k \leq \lfloor \log_2 N \rfloor} 2^{-krs} \mathrm{Tr}(A_s^{(\lfloor N/2^k \rfloor)})^r. \quad (\text{D.1})$$

The summands in equation (D.1) do not share a common truncation, so the finite sum is not a geometric factor. Only the analytic infinite-operator proof licenses equation (6.2).

The two evaluator sources have different digests, use no project-local imports, and share neither expanded fixtures nor serialized intermediates. The adversarial closeout exercised 62 mutations in 25 families across 162 designated consumer invocations, with no survivor. A separate frozen audit rejected all 13 physical mutated clones across 22 invocations. These are reproducibility and type-discipline metadata, not additional mathematical evidence.

The comparison type is `FINITE_EXACT_DIAGNOSTIC`; its recorded infinite-theorem status is `NOT_INFERRED_FROM_FINITE_EVIDENCE`. State A contains no publication manifest or State-B provenance, and this appendix makes no such claim. These checks establish reproducibility of the evaluated artifact only. They do not prove an endpoint, external priority, rational-prime emergence, or any target divisor.